

COMPASS_FE_20221220

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R Markdown

#notes I don't have a system yet to compare old slopes to current slopes for that day #I don't have a system for eliminating wells if the sd is high - if you think one of your sample wells was off you would need to remove it manually...

#Files Required: #csv of standards as list not plate with columns: Fe(II), Abs, Curve, FE ; these are: Known std concentration, Absorbance, Curve type (High or Low), and FE Curve (FE_2 or FE_Total)
#csv of samples as list not plate with columns: Date, Plate, Well_ID, SampleID, Dil_Factor, Abs1, Abs2 ; these are Date of run, Plate ID, Well ID, Sample ID, Dilution factor, Absorbance 1 (Fe2 or first read), and Absorbance 2 (feTot or second read) # I save these files in a folder for each day of FE runs

Read in Standards and subset

```
stds <- read.csv("Raw Data/FE_20221219_Std.csv")
head(stds)
```

```
##   Fe.II.      Abs Curve   FE
## 1    60 0.14050000   Low FE_2
## 2    45 0.12900000   Low FE_2
## 3    30 0.09050000   Low FE_2
## 4    18 0.06900000   Low FE_2
## 5     9 0.05500000   Low FE_2
## 6     6 0.04866667   Low FE_2
```

```
#subset by FE_2 and FE_Total
FE_2_std <- stds[stds$FE == "FE_2", ] # One factor level
head(FE_2_std)
```

```
##   Fe.II.      Abs Curve   FE
## 1    60 0.14050000   Low FE_2
## 2    45 0.12900000   Low FE_2
## 3    30 0.09050000   Low FE_2
## 4    18 0.06900000   Low FE_2
## 5     9 0.05500000   Low FE_2
## 6     6 0.04866667   Low FE_2
```

```
FE_Tot_stds <- stds[stds$FE == "FE_Total", ] # One factor level
head(FE_Tot_stds)
```

```
##      Fe.II.      Abs Curve      FE
## 31      60 0.31150000    Low FE_Total
## 32      45 0.23250000    Low FE_Total
## 33      30 0.17100000    Low FE_Total
## 34      18 0.11450000    Low FE_Total
## 35       9 0.08366667    Low FE_Total
## 36       6 0.06466667    Low FE_Total
```

#subset curves for high and low

```
FE_2_stds_high <- FE_2_stds[FE_2_stds$Curve == "High", ] # One factor level
head(FE_2_stds_high)
```

```
##      Fe.II.      Abs Curve      FE
## 19      500 1.3740000    High FE_2
## 20      250 0.7016667    High FE_2
## 21      100 0.2913333    High FE_2
## 22       60 0.1820000    High FE_2
## 23       45 0.1625000    High FE_2
## 24       30 0.1255000    High FE_2
```

```
FE_2_stds_low <- FE_2_stds[FE_2_stds$Curve == "Low", ] # One factor level
head(FE_2_stds_low)
```

```
##      Fe.II.      Abs Curve      FE
## 1       60 0.14050000    Low FE_2
## 2       45 0.12900000    Low FE_2
## 3       30 0.09050000    Low FE_2
## 4       18 0.06900000    Low FE_2
## 5        9 0.05500000    Low FE_2
## 6        6 0.04866667    Low FE_2
```

```
FE_Tot_stds_high <- FE_Tot_stds[FE_Tot_stds$Curve == "High", ] # One factor level
head(FE_Tot_stds_high)
```

```
##      Fe.II.      Abs Curve      FE
## 49      500 1.4390000    High FE_Total
## 50      250 0.8456667    High FE_Total
## 51      100 0.4793333    High FE_Total
## 52       60 0.2965000    High FE_Total
## 53       45 0.2345000    High FE_Total
## 54       30 0.1720000    High FE_Total
```

```
FE_Tot_stds_low <- FE_Tot_stds[FE_Tot_stds$Curve == "Low", ] # One factor level
head(FE_Tot_stds_low)
```

```
##      Fe.II.      Abs Curve      FE
## 31      60 0.31150000    Low FE_Total
```

```
## 32      45 0.23250000 Low FE_Total
## 33      30 0.17100000 Low FE_Total
## 34      18 0.11450000 Low FE_Total
## 35       9 0.08366667 Low FE_Total
## 36       6 0.06466667 Low FE_Total
```

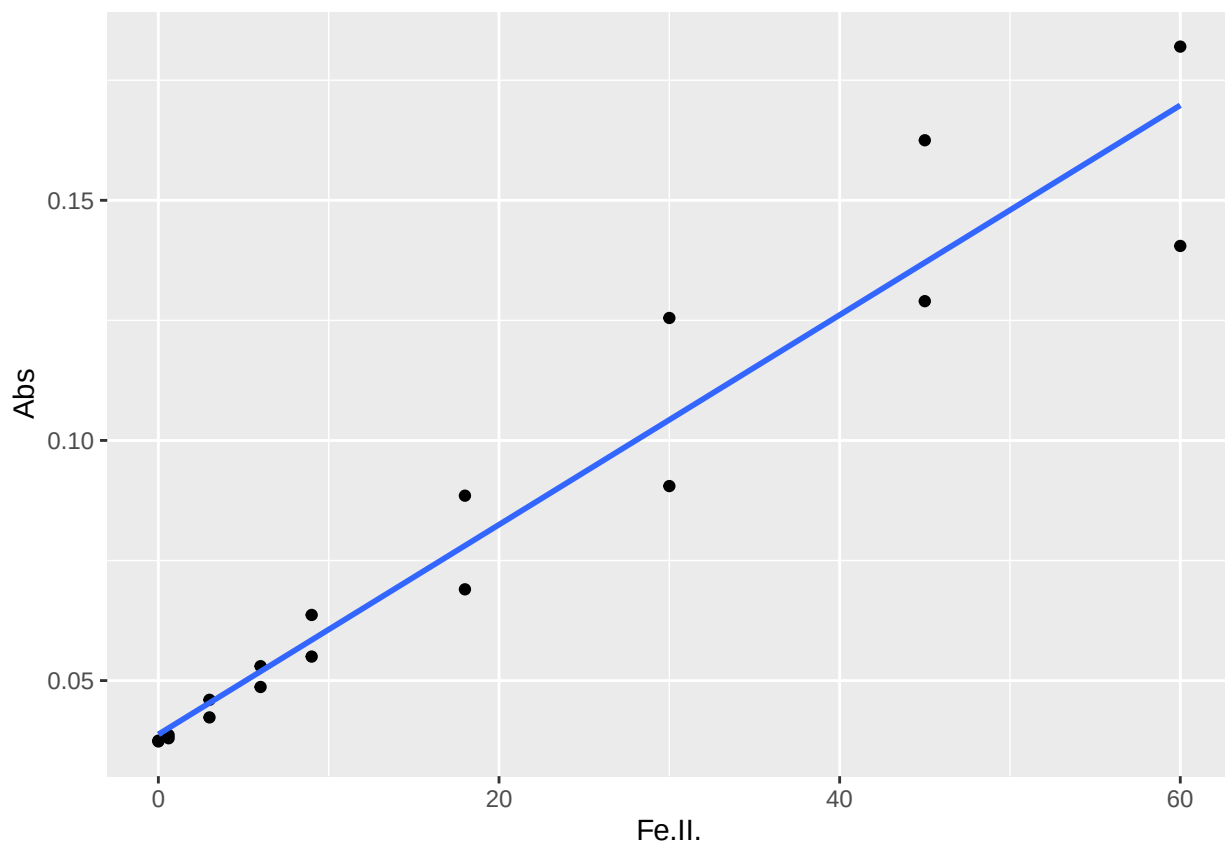
Plot standards

```
#Plot each of these and calculate the slope, intercept, and R2
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3
```

```
#Low curve for Fe(II)
FE_2_Low <- ggplot(FE_2_stds_low, aes(Fe.II., Abs)) + geom_point() + geom_smooth(method = "lm", se = FALSE)
FE_2_Low
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
FE_2_Low_lm <- lm(FE_2_stds_low$Abs ~ FE_2_stds_low$Fe.II.)
summary(FE_2_Low_lm)
```

```
##
## Call:
## lm(formula = FE_2_stds_low$Abs ~ FE_2_stds_low$Fe.II.)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.029282 -0.003397 -0.001459  0.004187  0.025462
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.0388040   0.0041404    9.372 6.72e-08 ***
## FE_2_stds_low$Fe.II. 0.0021830   0.0001487   14.678 1.06e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0128 on 16 degrees of freedom
## Multiple R-squared:  0.9309, Adjusted R-squared:  0.9265
## F-statistic: 215.4 on 1 and 16 DF,  p-value: 1.058e-10
```

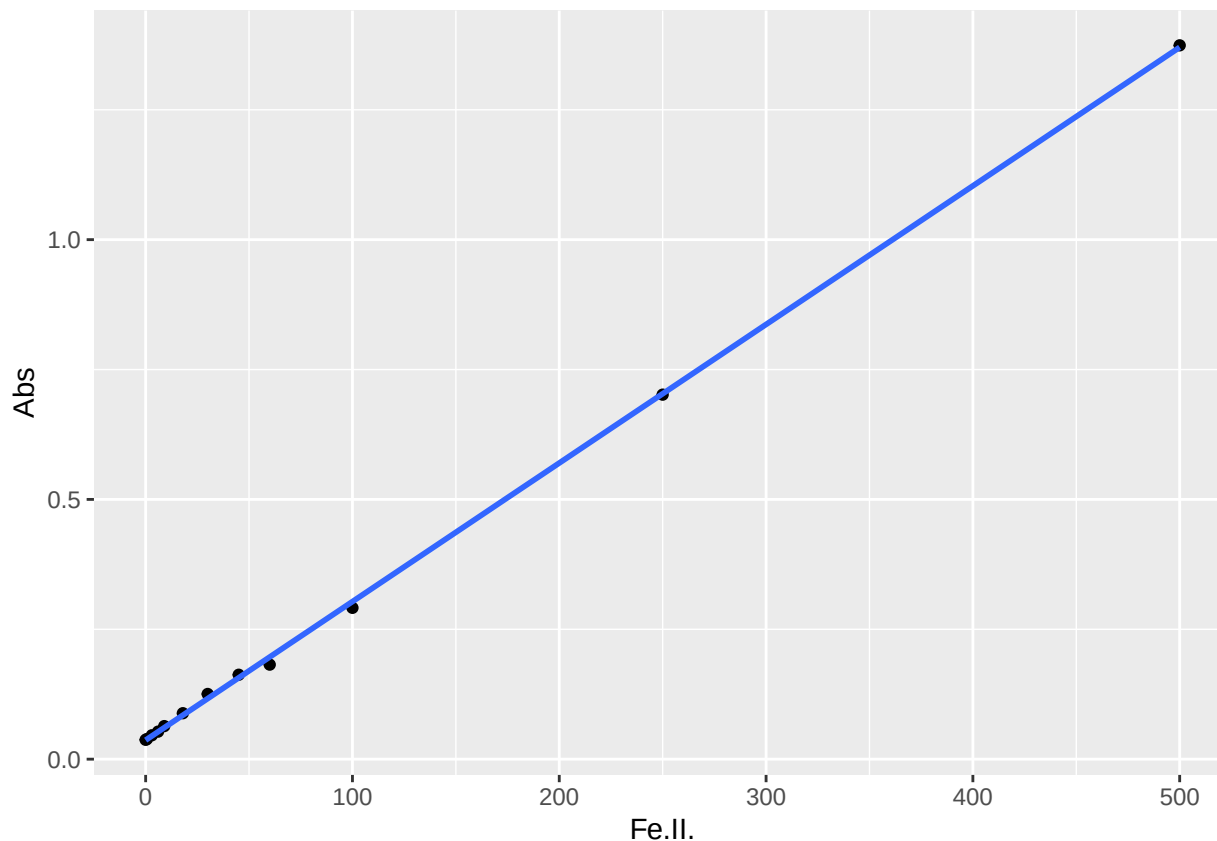
```
cf <- coef(FE_2_Low_lm)

#create data frame with 1 rows and 0 columns
Slopes1 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes1$Curve <- "FE_2_Low"
Slopes1$R2 <- summary(FE_2_Low_lm)$adj.r.squared
Slopes1$Slope <- cf[2]
Slopes1$Intercept <- cf[1]
head(Slopes1)
```

```
##      Curve      R2      Slope Intercept
## 1 FE_2_Low 0.9265477 0.002182967 0.03880403
```

```
#High curve for Fe(III)
FE_2_High <- ggplot(FE_2_stds_high, aes(Fe.II., Abs)) + geom_point() + geom_smooth(method = "lm", se = 1)
FE_2_High
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
FE_2_High_lm <- lm(FE_2_stds_high$Abs ~ FE_2_stds_high$Fe.II.)
summary(FE_2_High_lm)
```

```
##
## Call:
## lm(formula = FE_2_stds_high$Abs ~ FE_2_stds_high$Fe.II.)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-0.0146437	-0.0001477	0.0010572	0.0037948	0.0088739

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.661e-02	2.417e-03	15.14	3.19e-08 ***
FE_2_stds_high\$Fe.II.	2.667e-03	1.459e-05	182.84	< 2e-16 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.007184 on 10 degrees of freedom
## Multiple R-squared:  0.9997, Adjusted R-squared:  0.9997
## F-statistic: 3.343e+04 on 1 and 10 DF,  p-value: < 2.2e-16
```

```
cf <- coef(FE_2_High_lm)
```

```
#create data frame with 1 rows and 0 columns
```

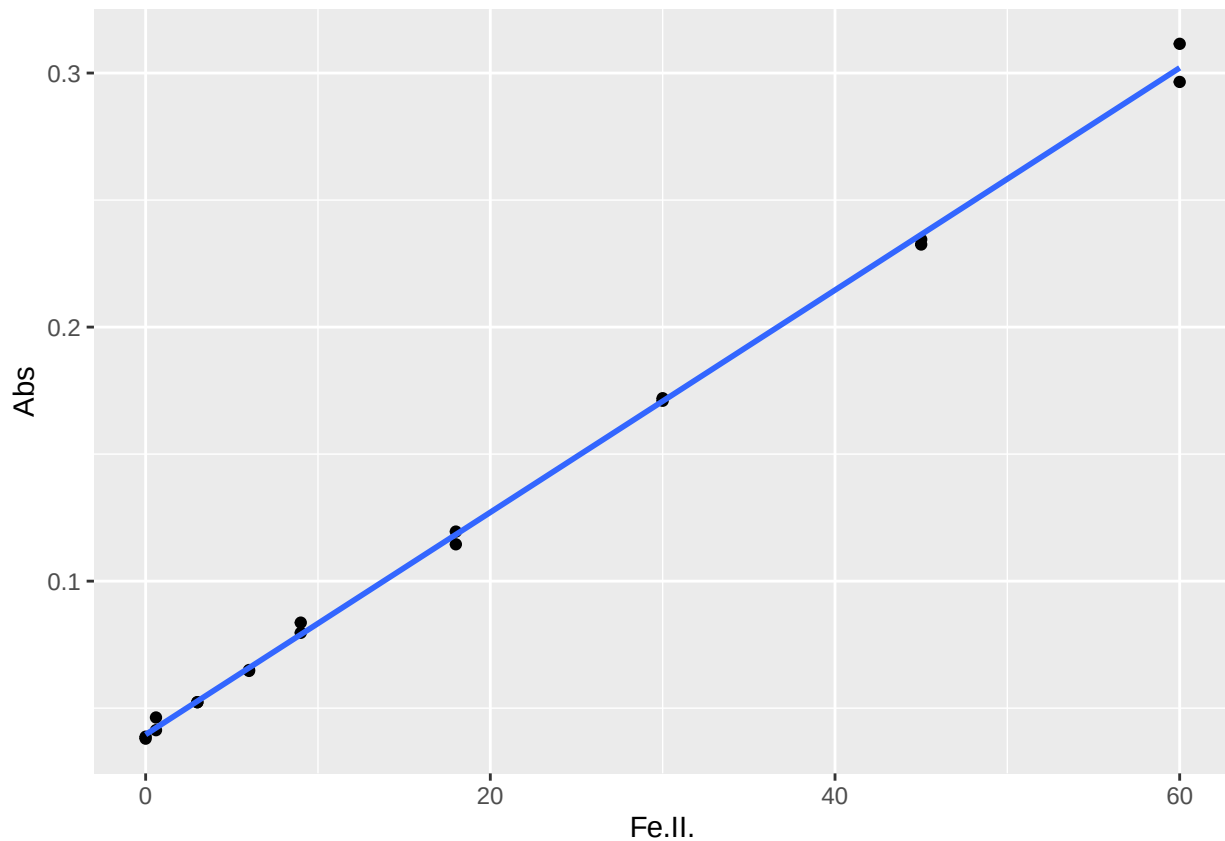
```
Slopes2 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes2$Curve <- "FE_2_High"
Slopes2$R2 <- summary(FE_2_High_lm)$adj.r.squared
Slopes2$Slope <- cf[2]
Slopes2$Intercept <- cf[1]
head(Slopes2)
```

```
##      Curve      R2      Slope Intercept
## 1 FE_2_High 0.9996711 0.002667251 0.03660863
```

```
#Low curve for Fe(Total)
```

```
FE_Tot_Low <- ggplot(FE_Tot_stds_low, aes(Fe.II., Abs)) + geom_point() + geom_smooth(method = "lm", se = FALSE)
FE_Tot_Low
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
FE_Tot_Low_lm <- lm(FE_Tot_stds_low$Abs ~ FE_Tot_stds_low$Fe.II.)
summary(FE_Tot_Low_lm)
```

```
##
## Call:
## lm(formula = FE_Tot_stds_low$Abs ~ FE_Tot_stds_low$Fe.II.)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0055361 -0.0015061 -0.0006273  0.0010432  0.0094639
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.961e-02  1.152e-03   34.4  <2e-16 ***
## FE_Tot_stds_low$Fe.II. 4.374e-03  4.136e-05  105.7  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.00356 on 16 degrees of freedom
## Multiple R-squared:  0.9986, Adjusted R-squared:  0.9985
## F-statistic: 1.118e+04 on 1 and 16 DF,  p-value: < 2.2e-16
```

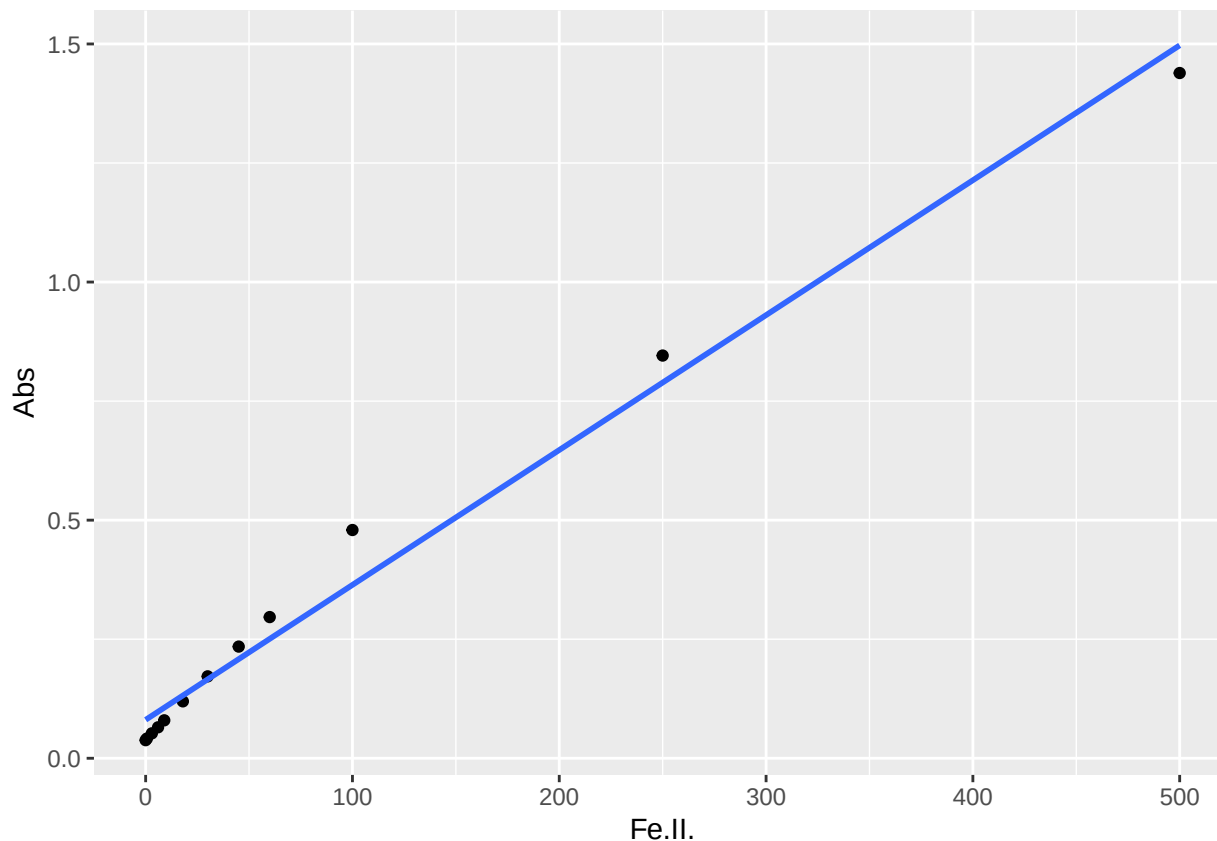
```
cf <- coef(FE_Tot_Low_lm)

#create data frame with 1 rows and 0 columns
Slopes3 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes3$Curve <- "FE_Tot_Low"
Slopes3$R2 <- summary(FE_Tot_Low_lm)$adj.r.squared
Slopes3$Slope <- cf[2]
Slopes3$Intercept <- cf[1]
head(Slopes3)
```

```
##      Curve      R2      Slope Intercept
## 1 FE_Tot_Low 0.9984817 0.004373732 0.03961213
```

```
#High curve for Fe(III)
FE_Tot_High <- ggplot(FE_Tot_stds_high, aes(Fe.II., Abs)) + geom_point() + geom_smooth(method = "lm", s
FE_Tot_High
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
FE_Tot_High_lm <- lm(FE_Tot_stds_high$Abs ~ FE_Tot_stds_high$Fe.II.)
summary(FE_Tot_High_lm)
```

```
##
## Call:
## lm(formula = FE_Tot_stds_high$Abs ~ FE_Tot_stds_high$Fe.II.)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.05809 -0.03796 -0.01940  0.03116  0.11532
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.0807464  0.0182844   4.416   0.0013 **
## FE_Tot_stds_high$Fe.II. 0.0028327  0.0001103  25.671 1.85e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05434 on 10 degrees of freedom
## Multiple R-squared:  0.9851, Adjusted R-squared:  0.9836
## F-statistic: 659 on 1 and 10 DF, p-value: 1.848e-10
```

```
cf <- coef(FE_Tot_High_lm)
```

```
#create data frame with 1 rows and 0 columns
```

```
Slopes4 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes4$Curve <- "FE_Tot_High"
Slopes4$R2 <- summary(FE_Tot_High_lm)$adj.r.squared
Slopes4$Slope <- cf[2]
Slopes4$Intercept <- cf[1]
head(Slopes4)
```

```
##           Curve          R2          Slope Intercept
## 1 FE_Tot_High 0.9835574 0.002832691 0.08074637
```

```
#pull all the slopes together into one dataframe
Slopes <- rbind(Slopes1, Slopes2, Slopes3, Slopes4)
head(Slopes)
```

```
##           Curve          R2          Slope Intercept
## 1   FE_2_Low 0.9265477 0.002182967 0.03880403
## 2   FE_2_High 0.9996711 0.002667251 0.03660863
## 3 FE_Tot_Low 0.9984817 0.004373732 0.03961213
## 4 FE_Tot_High 0.9835574 0.002832691 0.08074637
```

```
####
```

Read in Sample Data subset

```
dat <- read.csv("Raw Data/FE_20221219_Samples.csv")
head(dat)
```

```
##      Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes
## 1 20221219 Samples1      1      stds          1 0.221 0.334
## 2 20221219 Samples1      2      stds          1 0.174 0.323    BR
## 3 20221219 Samples1      3      stds          1 0.196 0.320
## 4 20221219 Samples1      4          1          1 0.050 0.056
## 5 20221219 Samples1      5          1          1 0.050 0.056
## 6 20221219 Samples1      6          1          1 0.057 0.057
```

```
## remove any lines that say stds in SampleID
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##      filter, lag
```

```
## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union
```

```
dat <- dat %>% filter(!SampleID == 'stds')
dat <- dat %>% filter(!Notes == 'BR')
dat <- dat %>% filter(!Notes == 'ND')
head(dat)
```

```
##      Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes
## 1 20221219 Samples1      4         1          1 0.050 0.056
## 2 20221219 Samples1      5         1          1 0.050 0.056
## 3 20221219 Samples1      6         1          1 0.057 0.057
## 4 20221219 Samples1     10        15          1 0.061 0.062
## 5 20221219 Samples1     11        15          1 0.059 0.059
## 6 20221219 Samples1     12        15          1 0.060 0.060
```

```
#Now flag any Abs1 that are above the 60uM standard absorbance (0.2)
dat$FE_2_Curve <- ifelse(dat$Abs1 >0.2, "High", "Low")
head(dat)
```

```
##      Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221219 Samples1      4         1          1 0.050 0.056      Low
## 2 20221219 Samples1      5         1          1 0.050 0.056      Low
## 3 20221219 Samples1      6         1          1 0.057 0.057      Low
## 4 20221219 Samples1     10        15          1 0.061 0.062      Low
## 5 20221219 Samples1     11        15          1 0.059 0.059      Low
## 6 20221219 Samples1     12        15          1 0.060 0.060      Low
```

```
#Now flag any Abs2 that are above the 60uM standard absorbance (0.3)
dat$FE_Tot_Curve <- ifelse(dat$Abs2 >0.3, "High", "Low")
head(dat)
```

```
##      Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221219 Samples1      4         1          1 0.050 0.056      Low
## 2 20221219 Samples1      5         1          1 0.050 0.056      Low
## 3 20221219 Samples1      6         1          1 0.057 0.057      Low
## 4 20221219 Samples1     10        15          1 0.061 0.062      Low
## 5 20221219 Samples1     11        15          1 0.059 0.059      Low
## 6 20221219 Samples1     12        15          1 0.060 0.060      Low
## FE_Tot_Curve
## 1      Low
## 2      Low
## 3      Low
## 4      Low
## 5      Low
## 6      Low
```

Subset Data and Calculate Concentrations

```
#Calculate concentrations of Fe(II)
dat$FE_2_Conc <- ifelse(dat$Abs1 >0.2, (dat$Abs1-Slopes2$Intercept)/Slopes2$Slope,
head(dat)
```

```
##      Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221219 Samples1      4         1          1 0.050 0.056          Low
## 2 20221219 Samples1      5         1          1 0.050 0.056          Low
## 3 20221219 Samples1      6         1          1 0.057 0.057          Low
## 4 20221219 Samples1     10        15          1 0.061 0.062          Low
## 5 20221219 Samples1     11        15          1 0.059 0.059          Low
## 6 20221219 Samples1     12        15          1 0.060 0.060          Low
##   FE_Tot_Curve FE_2_Conc
## 1           Low  5.128788
## 2           Low  5.128788
## 3           Low  8.335433
## 4           Low 10.167802
## 5           Low  9.251618
## 6           Low  9.709710
```

```
#calculate concentrations of Fe(total)
```

```
dat$FE_Tot_Conc <- ifelse(dat$Abs2 > 0.3, (dat$Abs2 - Slopes4$Intercept) / Slopes4$Slope,
head(dat))
```

```
##      Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221219 Samples1      4         1          1 0.050 0.056          Low
## 2 20221219 Samples1      5         1          1 0.050 0.056          Low
## 3 20221219 Samples1      6         1          1 0.057 0.057          Low
## 4 20221219 Samples1     10        15          1 0.061 0.062          Low
## 5 20221219 Samples1     11        15          1 0.059 0.059          Low
## 6 20221219 Samples1     12        15          1 0.060 0.060          Low
##   FE_Tot_Curve FE_2_Conc FE_Tot_Conc
## 1           Low  5.128788  3.746884
## 2           Low  5.128788  3.746884
## 3           Low  8.335433  3.975522
## 4           Low 10.167802  5.118710
## 5           Low  9.251618  4.432797
## 6           Low  9.709710  4.661435
```

```
#Account for the dilution factor
```

```
dat$FE_2_Conc_DilCorr <- dat$FE_2_Conc * dat$Dil_Factor
```

```
dat$FE_Tot_Conc_DilCorr <- dat$FE_Tot_Conc * dat$Dil_Factor
```

```
head(dat)
```

```
##      Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221219 Samples1      4         1          1 0.050 0.056          Low
## 2 20221219 Samples1      5         1          1 0.050 0.056          Low
## 3 20221219 Samples1      6         1          1 0.057 0.057          Low
## 4 20221219 Samples1     10        15          1 0.061 0.062          Low
## 5 20221219 Samples1     11        15          1 0.059 0.059          Low
## 6 20221219 Samples1     12        15          1 0.060 0.060          Low
##   FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr FE_Tot_Conc_DilCorr
## 1           Low  5.128788  3.746884          5.128788          3.746884
## 2           Low  5.128788  3.746884          5.128788          3.746884
## 3           Low  8.335433  3.975522          8.335433          3.975522
## 4           Low 10.167802  5.118710         10.167802          5.118710
```

```
## 5      Low  9.251618    4.432797      9.251618      4.432797
## 6      Low  9.709710    4.661435      9.709710      4.661435
```

Calculate Fe(III)

```
#Calculate concentrations of Fe(III)
dat$FE_3_Conc <- dat$FE_Tot_Conc_DilCorr-dat$FE_2_Conc_DilCorr

#Use ifelse to make any negative values equal to zero
dat$FE_2_Conc_Final <- ifelse(dat$FE_2_Conc_DilCorr <0, 0, dat$FE_2_Conc_DilCorr)

dat$FE_3_Conc_Final <- ifelse(dat$FE_3_Conc <0, 0, dat$FE_3_Conc)

#Use ifelse to make any negative values equal to zero
dat$FE_2_Info <- ifelse(dat$FE_2_Conc_DilCorr <0, 'bdl', 'within range')

#Use ifelse to make any negative values equal to zero
dat$FE_3_Info <- ifelse(dat$FE_3_Conc <0, 'bdl', 'within range')

head(dat)
```

```
##      Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221219 Samples1      4         1          1 0.050 0.056      Low
## 2 20221219 Samples1      5         1          1 0.050 0.056      Low
## 3 20221219 Samples1      6         1          1 0.057 0.057      Low
## 4 20221219 Samples1     10        15          1 0.061 0.062      Low
## 5 20221219 Samples1     11        15          1 0.059 0.059      Low
## 6 20221219 Samples1     12        15          1 0.060 0.060      Low
##      FE_Tot_Conc FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr FE_Tot_Conc_DilCorr
## 1      Low    5.128788    3.746884      5.128788      3.746884
## 2      Low    5.128788    3.746884      5.128788      3.746884
## 3      Low    8.335433    3.975522      8.335433      3.975522
## 4      Low   10.167802    5.118710     10.167802      5.118710
## 5      Low    9.251618    4.432797      9.251618      4.432797
## 6      Low    9.709710    4.661435      9.709710      4.661435
##      FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final      FE_2_Info FE_3_Info
## 1 -1.381904      5.128788          0 within range      bdl
## 2 -1.381904      5.128788          0 within range      bdl
## 3 -4.359911      8.335433          0 within range      bdl
## 4 -5.049092     10.167802          0 within range      bdl
## 5 -4.818820      9.251618          0 within range      bdl
## 6 -5.048275      9.709710          0 within range      bdl
```

Calculate Averages across wells and std. dev.

```
head(dat)
```

```
##      Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221219 Samples1      4         1          1 0.050 0.056      Low
```

```
## 2 20221219 Samples1      5      1      1 0.050 0.056      Low
## 3 20221219 Samples1      6      1      1 0.057 0.057      Low
## 4 20221219 Samples1     10     15      1 0.061 0.062      Low
## 5 20221219 Samples1     11     15      1 0.059 0.059      Low
## 6 20221219 Samples1     12     15      1 0.060 0.060      Low
##   FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr FE_Tot_Conc_DilCorr
## 1           Low  5.128788   3.746884      5.128788      3.746884
## 2           Low  5.128788   3.746884      5.128788      3.746884
## 3           Low  8.335433   3.975522      8.335433      3.975522
## 4           Low 10.167802   5.118710     10.167802      5.118710
## 5           Low  9.251618   4.432797      9.251618      4.432797
## 6           Low  9.709710   4.661435      9.709710      4.661435
##   FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final   FE_2_Info FE_3_Info
## 1 -1.381904      5.128788      0 within range      bdl
## 2 -1.381904      5.128788      0 within range      bdl
## 3 -4.359911      8.335433      0 within range      bdl
## 4 -5.049092     10.167802      0 within range      bdl
## 5 -4.818820      9.251618      0 within range      bdl
## 6 -5.048275      9.709710      0 within range      bdl
```

```
#summarize by sampleID so that we can calculate the mean and std. dev. of the three wells
dat1 <- dat %>%
  group_by(SampleID) %>%
  summarise(FE_2_mean = mean(FE_2_Conc_Final), FE_2_sd = sd(FE_2_Conc_Final),
            FE_3_mean = mean(FE_3_Conc_Final), FE_3_sd = sd(FE_3_Conc_Final),
            FE_2_flag = first(FE_2_Info), FE_3_flag = first(FE_3_Info))

head(dat1)
```

```
## # A tibble: 6 x 7
##   SampleID FE_2_mean FE_2_sd FE_3_mean FE_3_sd FE_2_flag   FE_3_flag
##   <chr>      <dbl>   <dbl>   <dbl>   <dbl> <chr>      <chr>
## 1 1         6.20    1.85      0      0 0 within range bdl
## 2 13        2.38     0        0      0 0 within range bdl
## 3 14        2.53    0.700     0      0 0 within range bdl
## 4 15        9.71    0.458     0      0 0 within range bdl
## 5 16        2.69    0.264     0      0 0 within range bdl
## 6 17        1.01     0        0      0 0 within range bdl
```

Export full data then just final data

```
#Read in meta-data and sample IDs
IDs <- read.csv("Raw Data/FE_SampleIDs_2022.csv")
head(IDs)
```

```
##      Date Month Year Campaign Site Zone Depth Rep
## 1 20220502   May 2022   MonMon  MSM   WC   10   A
## 2 20220502   May 2022   MonMon  MSM   WC   10   B
## 3 20220502   May 2022   MonMon  MSM   WC   10   C
## 4 20220502   May 2022   MonMon  MSM   WC   20   A
## 5 20220502   May 2022   MonMon  MSM   WC   20   B
```

```
## 6 20220502 May 2022 MonMon MSM WC 20 C
## Sample_Name SampleID
## 1 MSM_WC_PW_MonMon_20220502_SipA_10cm 654
## 2 MSM_WC_PW_MonMon_20220502_SipB_10cm 656
## 3 MSM_WC_PW_MonMon_20220502_SipC_10cm 658
## 4 MSM_WC_PW_MonMon_20220502_SipA_20cm 655
## 5 MSM_WC_PW_MonMon_20220502_SipB_20cm 657
## 6 MSM_WC_PW_MonMon_20220502_SipC_20cm 659
```

```
dat_IDs <- merge(IDs, dat1)
head(dat_IDs)
```

```
## SampleID Date Month Year Campaign Site Zone Depth Rep
## 1 1 202211 November 2022 MonMon Gcrew UP 10 B
## 2 2 202211 November 2022 MonMon Gcrew UP 20 B
## 3 3 202211 November 2022 MonMon Gcrew UP 10 C
## 4 4 202211 November 2022 MonMon Gcrew UP 20 C
## 5 6 202211 November 2022 MonMon Gcrew TR 45 A
## 6 7 202211 November 2022 MonMon Gcrew TR 10 B
## Sample_Name FE_2_mean FE_2_sd FE_3_mean FE_3_sd FE_2_flag
## 1 GCrew_UP_LysB_10cm 6.197670 1.8513575 0.00000 0.00000 within range
## 2 GCrew_UP_LysB_20cm 4.517998 0.5289593 0.00000 0.00000 within range
## 3 GCrew_UP_LysC_10cm 9.098920 0.9535949 0.00000 0.00000 within range
## 4 GCrew_UP_LysC_20cm 7.113854 0.5289593 0.00000 0.00000 within range
## 5 GCrew_TR_LysA_45cm 455.297079 24.9200579 99.56533 19.17871 within range
## 6 GCrew_TR_LysB_10cm 354.819078 33.6571059 76.19172 44.50079 within range
## FE_3_flag
## 1 bdl
## 2 bdl
## 3 bdl
## 4 bdl
## 5 within range
## 6 within range
```

```
#Read out the summarized data
head(dat_IDs)
```

```
## SampleID Date Month Year Campaign Site Zone Depth Rep
## 1 1 202211 November 2022 MonMon Gcrew UP 10 B
## 2 2 202211 November 2022 MonMon Gcrew UP 20 B
## 3 3 202211 November 2022 MonMon Gcrew UP 10 C
## 4 4 202211 November 2022 MonMon Gcrew UP 20 C
## 5 6 202211 November 2022 MonMon Gcrew TR 45 A
## 6 7 202211 November 2022 MonMon Gcrew TR 10 B
## Sample_Name FE_2_mean FE_2_sd FE_3_mean FE_3_sd FE_2_flag
## 1 GCrew_UP_LysB_10cm 6.197670 1.8513575 0.00000 0.00000 within range
## 2 GCrew_UP_LysB_20cm 4.517998 0.5289593 0.00000 0.00000 within range
## 3 GCrew_UP_LysC_10cm 9.098920 0.9535949 0.00000 0.00000 within range
## 4 GCrew_UP_LysC_20cm 7.113854 0.5289593 0.00000 0.00000 within range
## 5 GCrew_TR_LysA_45cm 455.297079 24.9200579 99.56533 19.17871 within range
## 6 GCrew_TR_LysB_10cm 354.819078 33.6571059 76.19172 44.50079 within range
## FE_3_flag
## 1 bdl
```

```
## 2          bdl
## 3          bdl
## 4          bdl
## 5 within range
## 6 within range
```

```
write.csv(dat_IDs, file="Processed Data/FE_2022119_Summary_Data.csv")
```

```
#Read out all the data in dat
head(dat)
```

```
##      Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221219 Samples1      4         1          1 0.050 0.056          Low
## 2 20221219 Samples1      5         1          1 0.050 0.056          Low
## 3 20221219 Samples1      6         1          1 0.057 0.057          Low
## 4 20221219 Samples1     10        15          1 0.061 0.062          Low
## 5 20221219 Samples1     11        15          1 0.059 0.059          Low
## 6 20221219 Samples1     12        15          1 0.060 0.060          Low
## FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr FE_Tot_Conc_DilCorr
## 1          Low  5.128788   3.746884          5.128788          3.746884
## 2          Low  5.128788   3.746884          5.128788          3.746884
## 3          Low  8.335433   3.975522          8.335433          3.975522
## 4          Low 10.167802   5.118710         10.167802          5.118710
## 5          Low  9.251618   4.432797          9.251618          4.432797
## 6          Low  9.709710   4.661435          9.709710          4.661435
## FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final FE_2_Info FE_3_Info
## 1 -1.381904          5.128788          0 within range          bdl
## 2 -1.381904          5.128788          0 within range          bdl
## 3 -4.359911          8.335433          0 within range          bdl
## 4 -5.049092         10.167802          0 within range          bdl
## 5 -4.818820          9.251618          0 within range          bdl
## 6 -5.048275          9.709710          0 within range          bdl
```

```
write.csv(dat, file="Processed Data/FE_20221219_Data.csv")
```

```
#Now take out the absorbance and stuff
head(dat)
```

```
##      Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221219 Samples1      4         1          1 0.050 0.056          Low
## 2 20221219 Samples1      5         1          1 0.050 0.056          Low
## 3 20221219 Samples1      6         1          1 0.057 0.057          Low
## 4 20221219 Samples1     10        15          1 0.061 0.062          Low
## 5 20221219 Samples1     11        15          1 0.059 0.059          Low
## 6 20221219 Samples1     12        15          1 0.060 0.060          Low
## FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr FE_Tot_Conc_DilCorr
## 1          Low  5.128788   3.746884          5.128788          3.746884
## 2          Low  5.128788   3.746884          5.128788          3.746884
## 3          Low  8.335433   3.975522          8.335433          3.975522
## 4          Low 10.167802   5.118710         10.167802          5.118710
## 5          Low  9.251618   4.432797          9.251618          4.432797
## 6          Low  9.709710   4.661435          9.709710          4.661435
```

```
##      FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final      FE_2_Info FE_3_Info
## 1 -1.381904      5.128788      0 within range      bdl
## 2 -1.381904      5.128788      0 within range      bdl
## 3 -4.359911      8.335433      0 within range      bdl
## 4 -5.049092     10.167802      0 within range      bdl
## 5 -4.818820      9.251618      0 within range      bdl
## 6 -5.048275      9.709710      0 within range      bdl
```

```
dat2 <- dat[ ,-(2:3)]
dat2 <- dat2[ ,-(3:12)]
head(dat2)
```

```
##      Date SampleID FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final      FE_2_Info
## 1 20221219      1 -1.381904      5.128788      0 within range
## 2 20221219      1 -1.381904      5.128788      0 within range
## 3 20221219      1 -4.359911      8.335433      0 within range
## 4 20221219     15 -5.049092     10.167802      0 within range
## 5 20221219     15 -4.818820      9.251618      0 within range
## 6 20221219     15 -5.048275      9.709710      0 within range
##      FE_3_Info
## 1      bdl
## 2      bdl
## 3      bdl
## 4      bdl
## 5      bdl
## 6      bdl
```

```
write.csv(dat2, file="Processed Data/FE_20221219_short_Data.csv")
```

END